

SOFTWARE ENGINEERING

Lab 4: VibeCoding

Problem Statement — Scenario 16

Hangman Challenge

Consolidated Lab Submission

Improve the supplied multi-round terminal Hangman game by fixing repeated guesses, integrating session statistics, adding difficulty-based scoring, and strengthening input validation.

Prepared by	Arin Singh
SRN	PES1UG24CS079
Course	UE24CS341A : Software Engineering

1. Submission Links

Before Video Link

<https://github.com/cosmocode-source/SE-Labs-PES1UG24CS079/blob/main/Lab-4/BeforeChangesVideo.mp4>

After Video Link

<https://github.com/cosmocode-source/SE-Labs-PES1UG24CS079/blob/main/Lab-4/AfterChangesVideo.mp4>

Prompt Evidence / Complete Chat History

<https://claude.ai/share/a3e4641d-0d52-4277-8891-91ec2c0f965c>

GitHub — Lab 4 Repository

<https://github.com/cosmocode-source/SE-Labs-PES1UG24CS079/tree/main/Lab-4>

2. Features / Bug Fixes Implemented

Task	Change	Expected Behaviour
1. Guess-state correctness	Check both correct and incorrect previous guesses.	A repeated letter does not remove another life or count as a new guess.
2. Session statistics	Integrate SessionStats to record completed rounds, wins, and best streak.	Statistics persist between rounds; a loss resets the current streak but not the best streak.
3. Difficulty and scoring	Add difficulty selection and a once-per-round hint deduction.	Difficulty controls lives and rewards; hint scoring remains consistent even when the score starts at zero.
4. Input and feedback	Validate guesses, commands, category choices, difficulty choices, and replay answers.	Invalid input leaves gameplay state unchanged and each action receives clear feedback.